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This week in FYRE Pathway 1, my partners and I worked on creating and testing custom-made electrical sensors using three different materials: silver paste, silver paint, and carbon paint. We started over at the Wilbur Powerhouse makerspace, using a laser cutter to shape a precise stencil of our sensor. After placing the stencil onto a glass slide, We spread our conductive material across it and used a spin coater to get a nice, even layer. Finally, we tested each material's resistance with a multimeter to measure its conductivity. The silver paste turned out to have the lowest resistance, while the carbon paint had the highest.

Overall, this lab really helped me grasp how selecting specific conductive materials directly affects performance, alongside teaching me how much patience is required during fabrication. My group also had to figure out how to troubleshoot issues whenever our experiments ran into hiccups. At one point, we mistakenly forgot a strip of tape on a glass substrate before heating it up, so we were forced to scrape away the damaged section and re-do the step. During a different test, removing our stencil ended up ripping the conductive pathway, which required us to carefully lay down fresh tape and apply additional coating to fix the patch. Working through these minor setbacks taught me how precise you must be throughout the process, which made seeing our final resistance results feel all the more satisfying.